MODELOS Y SERVICIOS DE DATOS [MYSD-MBDA]
SQL Básico
2026-1
Guia autoestudio 1/ 6

OBJETIVOS

Desarrollar competencias básicas para escribir consultas simples en SQL

- Consultas con proyecciones, restricciones y producto cruz
  `SELECT .. FROM .. WHERE ..`
- Dar nuevos nombres
  `AS`
- Resultados sin repeticiones
  `DISTINCT`
- Orden en el resultado de consulta
  `ORDER BY`
- Consultas que requieren agrupamiento
  `GROUP BY ... HAVING ...`
- Operadores para expresiones
  - Numéricos: `ABS, ROUND, FLOOR, CEIL, DIV, MOD`
  - Lógicos: `AND, OR, NOT`
  - De comparación: `=, !=, <>, >, <, >=, <= , BETWEEN`
  - Cadenas: `LEN, INSTR, SUBSTR, REPLACE, TRIM, CONCAT, LIKE (% _)`
  - Tiempo: `CURRENT_DATE, CURRENT_TIMESTAMP, EXTRACT, TO_CHAR`
  - Agrupamiento: `AVG, COUNT, MAX, MIN, SUM`
  - Condicionales: `CASE`
  - Cambio de tipo: `CAST`

ENTREGA

Publicar las respuestas en el espacio correspondiente en un archivo .zip , el nombre de este archivo debe ser la concatenación en orden alfabético de los primeros apellidos de cada uno de los miembros.

INVESTIGACION

Para que tener conocimiento suficiente sobre la tecnología que vamos a usar en el laboratorio, investiguen las siguientes preguntas. No olviden incluir la bibliografía.

A. SQL

- ➔ ¿Qué es? ¿Para qué sirve?
- ➔ ¿Qué es DML, DDL,DCL,TCL?
- ➔ En este autoestudio, ¿en qué escribimos? ¿por qué?

B. Motor de bases de datos y bases de datos

- ➔ ¿Qué son?
- ➔ ¿Qué motores ofrece sqlzoo.net [http://sqlzoo.net/]?
- ➔ ¿Qué bases de datos ofrece sqlzoo?

A. SQL

1. ¿Qué es? ¿Para qué sirve?

SQL (Structured Query Language) es un **lenguaje de programación especializado para gestionar y manipular datos dentro de sistemas de bases de datos relacionales**. Se utiliza para **comunicarse con la base de datos**, permitiendo consultar, insertar, actualizar, eliminar datos y controlar estructuras.
Sirve para **consultar información almacenada, modificar datos existentes, crear nuevas estructuras como tablas y administrar permisos de acceso** en la base de datos.

2. ¿Qué es DML, DDL, DCL, TCL?

Estas siglas representan **grupos de comandos dentro del lenguaje SQL**, que tienen diferentes propósitos:

DML (Data Manipulation Language):
Lenguaje para **manipular datos dentro de las tablas**, como añadir, borrar, actualizar o consultar datos.
Ejemplos: SELECT, INSERT, UPDATE, DELETE.

DDL (Data Definition Language):
Lenguaje para **definir o modificar la estructura de la base de datos**, como tablas o índices.
Ejemplos: CREATE, ALTER, DROP.

DCL (Data Control Language):
Lenguaje para **gestionar permisos de acceso y control de seguridad** dentro de la base de datos.
Ejemplos: GRANT, REVOKE.

TCL (Transaction Control Language):
Lenguaje para **controlar transacciones completas** (bloques de operaciones que deben ejecutarse juntas).
Ejemplos: COMMIT, ROLLBACK.

3. En este autoestudio, ¿en qué escribimos? ¿Por qué?

En este autoestudio escribimos **en SQL**, porque es el **lenguaje estándar para trabajar con bases de datos relacionales** y es compatible con la mayoría de los motores de bases de datos que existen (como MySQL, SQL Server, PostgreSQL, etc.).

B. Motor de bases de datos y bases de datos

1. ¿Qué son?

Motor de base de datos:
Es el **software interno que gestiona cómo se guardan, leen y manipulan los datos dentro de un sistema de base de datos**, ejecutando comandos SQL y manteniendo la integridad de la información.

Base de datos:
Es un **conjunto organizado de datos almacenados de forma estructurada**, que puede ser consultado y modificado eficientemente.

2. ¿Qué motores ofrece sqlzoo.net?

SQLZoo es una plataforma educativa para aprender SQL y ofrece ejercicios prácticos basados en varios **motores de bases de datos populares**, tales como:

- **SQL Server**
- **Oracle**
- **MySQL**
- **DB2**
- **PostgreSQL**

Esto permite practicar SQL con sintaxis real de distintos motores.

3. ¿Qué bases de datos ofrece sqlzoo?

SQLZoo proporciona **bases de datos de ejemplo con datos reales**, listas para practicar consultas. Entre las más usadas están:

- **World** – contiene datos de países y estadísticas.
- **Nobel** – contiene información de los ganadores del Premio Nobel.

Estas bases sirven para hacer ejercicios prácticos de SQL.

Bibliografía general

- **IBM: ¿Qué es SQL?**
  https://www.ibm.com/es-es/think/topics/structured-query-language
- **Wikipedia: Lenguaje de control de datos**
  https://es.wikipedia.org/wiki/Lenguaje_de_control_de_datos
- **Wikipedia: Motor de base de datos**
  https://es.wikipedia.org/wiki/Motor_de_base_de_datos
- **AWS: ¿Qué es una base de datos SQL?**
  https://aws.amazon.com/es/what-is/sql-database/
- **SQLZoo: Sitio oficial**
  https://sqlzoo.net/wiki/SQL_Zoo

PRACTICA

Usando SQLzoo.net [http://sqlzoo.net/]

[En auto01.doc]

A. Estudien las siguientes secciones y realicen todos los ejercicios propuestos.

[Escriban las respuestas a las preguntas que no las tienen]

Tutorials: Learn SQL step by step

0 SELECT basics
Some simple queries to get you started

1 SELECT name
Some pattern matching queries

2 SELECT from World
In which we query the World country profile table.

3 SELECT from Nobel
Additional practice of the basic features using a table of Nobel Prize winners.

4 SELECT within SELECT
In which we form queries using other queries.

5 SUM and COUNT
In which we apply aggregate functions. more the same

SELECT BASICS

1.

The example uses a WHERE clause to show the population of 'France'. Note that strings should be in 'single quotes';

Modify it to show the population of Germany

```
SELECT population FROM world
WHERE name = 'Germany';
```

Submit SQL restore default

Correct answer

population
80716000

Scandinavia

2.

Checking a list The word **IN** allows us to check if an item is in a list. The example shows the name and population for the countries 'Brazil', 'Russia', 'India' and 'China'.

Show the name and the population for 'Sweden', 'Norway' and 'Denmark'.

```
SELECT name, population FROM world
  WHERE name IN ('Sweden','Norway','Denmark');
```

Submit SQL restore default

Correct answer

name	population
Denmark	5634437
Norway	5124383
Sweden	9675885

Just the right size

3.

Which countries are not too small and not too big? `BETWEEN` allows range checking (range specified is inclusive of boundary values). The example below shows countries with an area of 250,000-300,000 sq. km. Modify it to show the country and the area for countries with an area between 200,000 and 250,000.

```
SELECT name, area FROM world
  WHERE area BETWEEN 200000 AND 250000
```

Submit SQL   restore default

Correct answer

name	area
Belarus	207600
Ghana	238533
Guinea	245857
Guyana	214969
Laos	236800
Romania	238391
Uganda	241550
United Kingdom	242900

SELECT names

1.

You can use `WHERE name LIKE 'B%'` to find the countries that start with "B".

- The % is a *wild-card* it can match any characters

Find the country that start with Y

```
SELECT name FROM world
  WHERE name LIKE 'Y%'
```

Submit SQL   restore default

Correct answer

name
Yemen

2.

Find the countries that end with y

```
SELECT name FROM world
  WHERE name LIKE '%Y'
```

Submit SQL   restore default

Correct answer

name
Turkey
Germany
Hungary
Italy
Norway
Vatican City
Paraguay
Uruguay

3.

Luxembourg has an **x** - so does one other country. List them both.

Find the countries that contain the letter x

```
SELECT name FROM world
  WHERE name LIKE '%X%'
```

Submit SQL   restore default

Correct answer

name
Luxembourg
Mexico

4.

Iceland, Switzerland end with **land** - but are there others?

Find the countries that end with land

```
SELECT name FROM world
  WHERE name LIKE '%land'
```

Submit SQL   restore default

Correct answer

name
Swaziland
Thailand
Finland
Iceland
Ireland
Poland
Switzerland
New Zealand

5.

Columbia starts with a **C** and ends with **ia** - there are two more like this.

Find the countries that start with C and end with ia

```
SELECT name FROM world
  WHERE name LIKE 'C%ia'
```

Submit SQL   restore default

Correct answer

name
Cambodia
Colombia
Croatia

6.

Greece has a double **e** - who has a double **o**?

Find the country that has oo in the name

```
SELECT name FROM world
  WHERE name LIKE '%oo%'
```

Submit SQL   restore default

Correct answer

name
Cameroon

7.

Bahamas has three **a** - who else?

Find the countries that have three or more a in the name

```
SELECT name FROM world
WHERE name LIKE '%a%a%a%'
```

Submit SQL   restore default

Correct answer

name
Central African Republic
Equatorial Guinea
Madagascar
Mauritania
Sao TomÃ© and PrÃncipe
Tanzania
Afghanistan

8.

India and Angola have an **n** as the second character. You can use the underscore as a single character wildcard.

```
SELECT name FROM world
 WHERE name LIKE '_n%'
ORDER BY name
```

Find the countries that have "t" as the second character.

```
SELECT name FROM world
 WHERE name LIKE '_t%'
ORDER BY name
```

Submit SQL   restore default

Correct answer

name
Ethiopia
Italy

9.

Les**o**th**o** and M**o**ld**o**va both have two o characters separated by two other characters.

Find the countries that have two "o" characters separated by two others.

```
SELECT name FROM world
 WHERE name LIKE '%_o__o%'
```

Submit SQL   restore default

Correct answer

name
Congo, Democratic Republic of
Congo, Republic of
Lesotho
Morocco
Sao TomÃ© and PrÃncipe
Mongolia
Moldova

10.

Cuba and Togo have four characters names.

Find the countries that have exactly four characters.

```
SELECT name FROM world
 WHERE name LIKE '____'
```

Submit SQL   restore default

name
Chad
Mali
Togo
Iran
Iraq
Laos
Oman
Cuba
Fiji
Peru

11.

The capital of **Luxembourg** is **Luxembourg**. Show all the countries where the capital is the same as the name of the country

Find the country where the name is the capital city.

```
SELECT name
FROM world
WHERE name = capital
```

Submit SQL   restore default

Correct answer

name
Djibouti
Luxembourg
San Marino
Singapore

12.

The capital of **Mexico** is **Mexico City**. Show all the countries where the capital has the country together with the word "City".

Find the country where the capital is the country plus "City".

The concat function

```
SELECT name
FROM world
 WHERE capital LIKE '%City'
```

Submit SQL   restore default

Correct answer

name
Guatemala
Kuwait
Mexico
Panama

13.

Find the capital and the name where the capital includes the name of the country.

```
SELECT capital,name
FROM world
WHERE capital LIKE concat('%',name,'%')
```

Submit SQL   restore default

Correct answer

capital	name
Andorra la Vella	Andorra
Djibouti	Djibouti
Guatemala City	Guatemala
Kuwait City	Kuwait
Luxembourg	Luxembourg
Mexico City	Mexico
Monaco-Ville	Monaco

14.

Find the capital and the name where the capital is an extension of name of the country.

You *should* include **Mexico City** as it is longer than **Mexico**. You *should not* include **Luxembourg** as the capital is the same as the country.

```
SELECT capital, name
FROM world
WHERE capital LIKE concat('%',name,'%') AND capital != name
```

Submit SQL   restore default

Correct answer

capital	name
Andorra la Vella	Andorra
Guatemala City	Guatemala
Kuwait City	Kuwait
Mexico City	Mexico
Monaco-Ville	Monaco
Panama City	Panama

15.

The capital of Monaco is **Monaco-Ville**: this is the name **Monaco** and the extension is **-Ville**.

Show the name and the extension where the capital is a proper (non-empty) extension of name of the country.

You can use the SQL function REPLACE.

```
SELECT name, REPLACE(capital,name,'')
FROM world
WHERE capital LIKE concat("%",REPLACE(name,capital,''),"%") AND
 REPLACE(capital,name,'') != ''
```

Submit SQL   restore default

Correct answer

name	REPLACE(capital,name,'')
Andorra	la Vella
Guatemala	City
Kuwait	City
Mexico	City
Monaco	-Ville
Panama	City

SELECT from WORLD Tutorial

Introduction

1.

Read the notes about this table. Observe the result of running this SQL command to show the name, continent and population of all countries.

```
SELECT name, continent, population FROM world
```

Submit SQL   restore default

Correct answer

name	continent	population
Afghanistan	Asia	25500100
Albania	Europe	2821977
Algeria	Africa	38700000
Andorra	Europe	76098
Angola	Africa	19183590
Antigua and Barbuda	Caribbean	86295
Argentina	South America	42669500

Large Countries

2.

How to use WHERE to filter records. Show the name for the countries that have a population of at least 200 million. 200 million is 200000000, there are eight zeros.

```
SELECT name FROM world
WHERE population >= 200000000
```

Submit SQL   restore default

Correct answer

name
Brazil
China
India
Indonesia
United States

Per capita GDP

3.

Give the `name` and the **per capita GDP** for those countries with a `population` of at least 200 million.

HELP:How to calculate per capita GDP

```
SELECT name, gdp/population FROM world
WHERE population >= 200000000
```

Submit SQL   restore default

Correct answer

name	gdp/population
Brazil	11115.2648
China	6121.7106
India	1504.7931
Indonesia	3482.0205
United States	51032.2945

South America In millions

4.

Show the `name` and `population` in millions for the countries of the `continent` 'South America'. Divide the population by 1000000 to get population in millions.

```
SELECT name, population/1000000 FROM world
WHERE continent = 'South America'
```

Submit SQL   restore default

name	population/1000000
Argentina	42.6695
Bolivia	10.0273
Brazil	202.7940
Chile	17.7730
Colombia	47.6620
Ecuador	15.7742
Guyana	0.7849
Paraguay	6.7834
Peru	30.4751
Saint Vincent and the Grenadines	0.1090

France, Germany, Italy

5.

Show the `name` and `population` for France, Germany, Italy

```
SELECT name, population FROM world
WHERE name IN ('France', 'Germany', 'Italy');
```

Submit SQL   restore default

Correct answer

name	population
France	65906000
Germany	80716000
Italy	60782668

United

6.

Show the countries which have a `name` that includes the word 'United'

```
SELECT name FROM world
WHERE name LIKE '%United%'
```

Submit SQL   restore default

Correct answer

name
United Arab Emirates
United Kingdom
United States

Two ways to be big

7.

Two ways to be big: A country is **big** if it has an area of more than 3 million sq km or it has a population of more than 250 million.

Show the countries that are big by area or big by population. Show name, population and area.

```
SELECT name,population,area FROM world
WHERE area >= 3000000 OR population >= 250000000
```

Submit SQL   restore default

Correct answer

name	population	area
Australia	23545500	7692024
Brazil	202794000	8515767
Canada	35427524	9984670
China	1365370000	9596961
India	1246160000	3166414
Indonesia	252164800	1904569
Russia	146000000	17125242

One or the other (but not both)

8.

Exclusive OR (XOR). Show the countries that are big by area (more than 3 million) or big by population (more than 250 million) but not both. Show name, population and area.

- Australia has a big area but a small population, it should be **included**.
- Indonesia has a big population but a small area, it should be **included**.
- China has a big population **and** big area, it should be **excluded**.
- United Kingdom has a small population and a small area, it should be **excluded**.

```
SELECT name,population,area FROM world
WHERE area >= 3000000 AND population < 250000000 OR (area <  3000000 AND
population >= 250000000)
```

Submit SQL   restore default

Correct answer

name	population	area
Australia	23545500	7692024
Brazil	202794000	8515767
Canada	35427524	9984670
Indonesia	252164800	1904569
Russia	146000000	17125242

Rounding

9.

Show the `name` and `population` in millions and the GDP in billions for the countries of the `continent` 'South America'. Use the ROUND function to show the values to two decimal places.

For Americas show population in millions and GDP in billions both to 2 decimal places.

Millions and billions

Missing decimals

```
SELECT name, ROUND(population/1000000,2),ROUND(gdp/1000000000,2)  FROM world
WHERE continent = 'South America'
```

Submit SQL   restore default

Correct answer

name	ROUND(population/1000000,2)	ROUND(gdp/1000000000,
Argentina	42.67	477.
Bolivia	10.03	27.
Brazil	202.79	2254.
Chile	17.77	268.
Colombia	47.66	369.
Ecuador	15.77	87.

Trillion dollar economies

10.

Show the `name` and per-capita GDP for those countries with a GDP of at least one trillion (1000000000000; that is 12 zeros). Round this value to the nearest 1000.

Show per-capita GDP for the trillion dollar countries to the nearest $1000.

```
SELECT name, ROUND(gdp/population,-3) FROM world
WHERE gdp >= 1000000000000
```

Submit SQL   restore default

Correct answer

name	ROUND(gdp/population,-3)
Australia	66000
Brazil	11000
Canada	45000
China	6000
France	40000
Germany	42000
India	2000

Name and capital have the same length

11.

Greece has capital Athens.

Each of the strings 'Greece', and 'Athens' has 6 characters.

Show the name and capital where the name and the capital have the same number of characters.

- You can use the LENGTH function to find the number of characters in a string

For Microsoft SQL Server the function LENGTH is LEN

```
SELECT name, capital FROM world
 WHERE LENGTH(name) = LENGTH(capital)
```

Submit SQL   restore default

Correct answer

name	capital
Algeria	Algiers
Angola	Luanda
Armenia	Yerevan
Botswana	Gaborone
Canada	Ottowa
Djibouti	Djibouti
Egypt	Cairo

Matching name and capital

12.

The capital of Sweden is Stockholm. Both words start with the letter 'S'.

Show the name and the capital where the first letters of each match. Don't include countries where the name and the capital are the same word.

- You can use the function LEFT to isolate the first character.
- You can use `<>` as the **NOT EQUALS** operator.

```
SELECT name, capital FROM world
WHERE LEFT(name,1) = LEFT(capital,1) AND name != capital
```

Submit SQL   restore default

Correct answer

name	capital
Algeria	Algiers
Andorra	Andorra la Vella
Barbados	Bridgetown
Belize	Belmopan
Brazil	BrasÃlia
Brunei	Bandar Seri Begawan
Burundi	Bujumbura

All the vowels

13.

Equatorial Guinea and **Dominican Republic** have all of the vowels (a e i o u) in the name. They don't count because they have more than one word in the name.

Find the country that has all the vowels and no spaces in its name.

- You can use the phrase `name NOT LIKE '%a%'` to exclude characters from your results.
- The query shown misses countries like Bahamas and Belarus because they contain at least one 'a'

```
SELECT name
FROM world
WHERE name LIKE '%A%'
AND name LIKE '%E%'
AND name LIKE '%I%'
AND name LIKE '%O%'
AND name LIKE '%U%'
AND name LIKE '%a%'
AND name LIKE '%e%'
AND name LIKE '%i%'
AND name LIKE '%o%'
AND name LIKE '%u%'
AND name NOT LIKE '% %'
```

Submit SQL restore default

Correct answer

name
Mozambique

SELECT from Nobel Tutorial

Winners from 1950

1.

Change the query shown so that it displays Nobel prizes for 1950.

```
SELECT yr, subject, winner
  FROM nobel
 WHERE yr = 1950
```

Submit SQL restore default

Correct answer

yr	subject	winner
1950	chemistry	Kurt Alder
1950	chemistry	Otto Diels
1950	literature	Bertrand Russell
1950	medicine	Edward Kendall
1950	medicine	Philip Hench
1950	medicine	Tadeus Reichstein
1950	peace	Ralph Bunche
1950	physics	Cecil Powell

1962 Literature

2.

Show who won the 1962 prize for literature.

```
SELECT winner
  FROM nobel
 WHERE yr = 1962
   AND subject = 'literature'
```

Submit SQL restore default

Correct answer

winner
John Steinbeck

Albert Einstein

3.

Show the year and subject that won 'Albert Einstein' his prize.

```
SELECT yr, subject
FROM nobel
WHERE winner = 'Albert Einstein'
```

Submit SQL

restore default

Correct answer

yr	subject
1921	physics

Recent Peace Prizes

4.

Give the name of the 'peace' winners since the year 2000, including 2000.

```
SELECT winner
FROM nobel
WHERE yr >= 2000 and subject = 'peace'
```

Submit SQL

restore default

Correct answer

winner
Kim Dae-jung
Kofi Annan
United Nations
Jimmy Carter
Shirin Ebadi
Wangari Maathai
International Atomic Energy Agency
Mohamed ElBaradei

Literature in the 1980's

5.

Show all details (**yr**, **subject**, **winner**) of the literature prize winners for 1980 to 1989 inclusive.

```
SELECT yr, subject, winner FROM nobel
WHERE subject = 'literature' AND yr BETWEEN 1980 AND 1989
```

Submit SQL

restore default

Correct answer

yr	subject	winner
1980	literature	Czes?aw Mi?osz
1981	literature	Elias Canetti
1982	literature	Gabriel García Márquez
1983	literature	William Golding
1984	literature	Jaroslav Seifert
1985	literature	Claude Simon
1986	literature	Wole Soyinka
1987	literature	Joseph Brodsky
1988	literature	Naguib Mahfouz
1989	literature	Camilo José Cela

Only Presidents

6.

Show all details of the presidential winners:

- Theodore Roosevelt
- Thomas Woodrow Wilson
- Jimmy Carter
- Barack Obama

```
SELECT * FROM nobel
 WHERE winner IN ('Theodore Roosevelt','Thomas Woodrow  Wilson','Jimmy
Carter','Barack Obama')
```

Submit SQL

restore default

Correct answer

yr	subject	winner
1906	peace	Theodore Roosevelt
2002	peace	Jimmy Carter
2009	peace	Barack Obama

John

7.

Show the winners with first name John

```
SELECT winner
FROM nobel
WHERE winner LIKE 'John%'
```

Submit SQL

restore default

Correct answer

winner
John Macleod
John Galsworthy
John Northrop
John Mott
John Cockcroft
John Enders
John Bardeen

Chemistry and Physics from different years

8.

Show the year, subject, and name of physics winners for 1980 together with the chemistry winners for 1984.

```
SELECT yr, subject, winner FROM nobel
WHERE (subject = 'physics' AND yr = 1980) OR (subject = 'chemistry' AND yr =
1984)
```

Submit SQL

restore default

Correct answer

yr	subject	winner
1980	physics	James Cronin
1980	physics	Val Fitch
1984	chemistry	Bruce Merrifield

Exclude Chemists and Medics

9.

Show the year, subject, and name of winners for 1980 excluding chemistry and medicine

```
SELECT * FROM nobel
WHERE yr = 1980 AND subject NOT IN ('chemistry','medicine')
```

Submit SQL   restore default

Correct answer

yr	subject	winner
1980	literature	Czes?aw Mi?osz
1980	peace	Adolfo Pérez Esquivel
1980	physics	James Cronin
1980	physics	Val Fitch

Early Medicine, Late Literature

10.

Show year, subject, and name of people who won a 'Medicine' prize in an early year (before 1910, not including 1910) together with winners of a 'Literature' prize in a later year (after 2004, including 2004)

```
SELECT * FROM nobel
WHERE (subject = 'Medicine' AND yr < 1910) OR (subject = 'Literature' AND yr >=
2004)
```

Submit SQL   restore default

Correct answer

yr	subject	winner
1901	medicine	Emil von Behring
1902	medicine	Ronald Ross
1903	medicine	Niels Ryberg Finsen
1904	medicine	Ivan Pavlov
1905	medicine	Robert Koch
1906	medicine	Camillo Golgi
1906	medicine	Santiago Ramón y Cajal

Umlaut

11.

Find all details of the prize won by PETER GRÜNBERG

Non-ASCII characters

```
SELECT * FROM nobel
WHERE winner = 'PETER GRÜNBERG'
```

Submit SQL   restore default

Correct answer

yr	subject	winner
2007	physics	Peter Grünberg

Apostrophe

12.

Find all details of the prize won by EUGENE O'NEILL

Escaping single quotes

```
SELECT * FROM nobel
WHERE winner = "EUGENE O'NEILL"
```

Submit SQL   restore default

Correct answer

yr	subject	winner
1936	literature	Eugene O'Neill

Knights of the realm

13.

Knights in order

List the winners, year and subject where the winner starts with Sir. Show the the most recent first, then by name order.

```
SELECT winner,yr,subject FROM nobel
WHERE winner LIKE 'Sir%'
ORDER BY yr DESC,winner
```

Submit SQL   restore default

Correct answer

winner	yr	subject
Sir Peter Ratcliffe	2019	medicine
Sir Gregory Winter	2018	chemistry
Sir Fraser Stoddart	2016	chemistry
Sir John Gurdon	2012	medicine
Sir Martin Evans	2007	medicine
Sir Peter Mansfield	2003	medicine
Sir Paul Nurse	2001	medicine

Chemistry and Physics last

14.

The expression **subject IN ('chemistry','physics')** can be used as a value - it will be **0** or **1**.

Show the 1984 winners and subject ordered by subject and winner name; but list chemistry and physics last.


```
SELECT winner, subject
  FROM nobel
 WHERE yr=1984
 ORDER BY subject IN ('physics','chemistry'),subject,winner
```

Submit SQL   restore default

Correct answer

winner	subject
Jaroslav Seifert	literature
César Milstein	medicine
GeorgesF. Köhler	medicine
Niels Jerne	medicine
Desmond Tutu	peace
Bruce Merrifield	chemistry
Carlo Rubbia	physics

SELECT within SELECT Tutorial

Bigger than Russia

1.

List each country name where the population is larger than that of 'Russia'.

```
world(name, continent, area, population, gdp)
```

```
SELECT name FROM world
  WHERE population >
     (SELECT population FROM world
      WHERE name='Russia')
```

Submit SQL   restore default

Correct answer

name
Bangladesh
Brazil
China
India
Indonesia
Nigeria
Pakistan
United States

Richer than UK

2.

Show the countries in Europe with a per capita GDP greater than 'United Kingdom'.

Per Capita GDP

```
SELECT name
  FROM world
    WHERE continent = 'Europe' AND gdp/population >
       (SELECT gdp/population
        FROM world
        WHERE name ='United Kingdom');
```

Submit SQL   restore default

Correct answer

name
Andorra
Austria
Belgium
Denmark
Finland
France
Germany

Neighbours of Argentina and Australia

3.

List the name and continent of countries in the continents containing either Argentina or Australia. Order by name of the country.

```
SELECT name, continent FROM world
WHERE continent IN (SELECT continent FROM world WHERE name IN
('Argentina','Australia'))
ORDER BY name
```

Submit SQL   restore default

Correct answer

name	continent
Argentina	South America
Australia	Oceania
Bolivia	South America
Brazil	South America
Chile	South America
Colombia	South America
Ecuador	South America
Fiji	Oceania
Guyana	South America

Between Canada and Poland

4.

Which country has a population that is more than United Kingdom but less than Germany? Show the name and the population.

```
SELECT name,population FROM world
WHERE population > (SELECT population FROM world WHERE name = "United Kingdom")
AND population < (SELECT population FROM world WHERE name = "Germany")
```

Submit SQL   restore default

Correct answer

name	population
Congo, Democratic Republic of	69360000
France	65906000
Iran	77552000
Thailand	64456700
Turkey	76667864

Percentages of Germany

5.

Germany (population roughly 80 million) has the largest population of the countries in Europe. Austria (population 8.5 million) has 11% of the population of Germany.

Show the name and the population of each country in Europe. Show the population as a percentage of the population of Germany.

The format should be *Name*, *Percentage* for example:

name	percentage
Albania	3%
Andorra	0%
Austria	11%
...	...

Decimal places

Percent symbol %

```
SELECT name, CONCAT(CAST(ROUND(100*population/(SELECT population FROM world
WHERE name = 'Germany'),0) as int), '%') AS percentage
FROM world
WHERE continent = 'Europe'
```

Submit SQL   restore default

Correct answer

name	percentage
Albania	3%
Andorra	0%
Austria	11%
Belarus	12%
Belgium	14%
Bosnia and Herzegovina	5%
Bulgaria	9%
Croatia	5%
Czech Republic	13%

Bigger than every country in Europe

6.

Which countries have a GDP greater than every country in Europe? [Give the name only.] (Some countries may have NULL gdp values)

```
SELECT name FROM world
WHERE gdp > ALL(SELECT gdp FROM world
         WHERE gdp>0 AND continent = 'Europe')
```

Submit SQL   restore default

Correct answer

name
China
Japan
United States

Largest in each continent

7.

Find the largest country (by area) in each continent, show the continent, the name and the area:

The above example is known as a **correlated** or **synchronized** sub-query.

Using correlated subqueries

```
SELECT continent, name, area FROM world x
  WHERE area >= ALL
    (SELECT area FROM world y
        WHERE y.continent=x.continent
          AND area > 0)
```

Submit SQL   restore default

Correct answer

continent	name	area
Africa	Algeria	2381741
Oceania	Australia	7692024
South America	Brazil	8515767
North America	Canada	9984670
Asia	China	9596961
Caribbean	Cuba	109884
Europe	Kazakhstan	2724900

First country of each continent (alphabetically)

8.

List each continent and the name of the country that comes first alphabetically.

```
Select  x.continent, x.name
From world x
Where x.name <= ALL (select y.name from world y where x.continent=y.continent)
ORDER BY continent
```

Submit SQL   restore default

Correct answer

continent	name
Africa	Algeria
Asia	Afghanistan
Caribbean	Antigua and Barbuda
Eurasia	Armenia
Europe	Albania
North America	Belize
Oceania	Australia

Difficult Questions That Utilize Techniques Not Covered In Prior Sections

9.

Find the continents where all countries have a population <= 25000000. Then find the names of the countries associated with these continents. Show name, continent and population.

```
SELECT name,continent,population FROM world x
WHERE 25000000 >= ALL(SELECT population FROM world y WHERE
x.continent=y.continent)
```

Submit SQL   restore default

Correct answer

name	continent	population
Antigua and Barbuda	Caribbean	86295
Australia	Oceania	23545500
Bahamas	Caribbean	351461
Barbados	Caribbean	285000
Cuba	Caribbean	11167325
Dominica	Caribbean	71293
Dominican Republic	Caribbean	9445281

Three time bigger

10.

Some countries have populations more than three times that of all of their neighbours (in the same continent). Give the countries and continents.

```
SELECT name, continent FROM world x
WHERE population/3 >= ALL(SELECT population FROM world y WHERE
    x.continent = y.continent AND x.name != y.name)
```

Submit SQL restore default

Correct answer

name	continent
Australia	Oceania
Brazil	South America
Russia	Eurasia

SUM and COUNT

Total world population

1.

Show the total **population** of the world.

```
world(name, continent, area, population, gdp)
```

```
SELECT SUM(population)
FROM world
```

Submit SQL restore default

Correct answer

SUM(population)
7118632738

List of continents

2.

List all the continents - just once each.

```
SELECT DISTINCT continent FROM world
```

Submit SQL restore default

Correct answer

continent
Africa
Asia
Caribbean
Eurasia
Europe
North America
Oceania

GDP of Africa

3.

Give the total GDP of Africa

```
SELECT SUM(gdp)
FROM world
WHERE continent = "Africa"
```

Submit SQL

restore default

Correct answer

SUM(gdp)
1811788000000

Count the big countries

4.

How many countries have an **area** of at least 1000000

```
SELECT COUNT(name) FROM world
WHERE area >= 1000000
```

Submit SQL

restore default

Correct answer

COUNT(name)
29

Baltic states population

5.

What is the total **population** of ('Estonia', 'Latvia', 'Lithuania')

```
SELECT SUM(population) FROM world
WHERE name IN ('Estonia', 'Latvia', 'Lithuania')
```

Submit SQL

restore default

Correct answer

SUM(population)
6251750

Using GROUP BY and HAVING

You may want to look at these examples: Using GROUP BY and HAVING.

Counting the countries of each continent

6.

For each **continent** show the **continent** and number of countries.

```
SELECT continent, COUNT(name) FROM world
GROUP BY continent
```

Submit SQL   restore default

Correct answer

continent	COUNT(name)
Africa	53
Asia	47
Caribbean	11
Eurasia	2
Europe	44
North America	11
Oceania	14

Counting big countries in each continent

7.

For each **continent** show the **continent** and number of countries with populations of at least 10 million.

```
SELECT continent, COUNT(name) FROM world
WHERE population >= 10000000
GROUP BY continent
```

Submit SQL   restore default

Correct answer

continent	COUNT(name)
Africa	29
Asia	26
Caribbean	2
Eurasia	1
Europe	14
North America	4
Oceania	1

Counting big continents

8.

List the continents that **have** a total population of at least 100 million.

```
SELECT continent
FROM world
GROUP BY continent
HAVING SUM(population) >= 100000000
```

Submit SQL   restore default

Correct answer

continent
Africa
Asia
Eurasia
Europe
North America
South America

in which we apply aggregate functions. more the same

B. Seleccionen cinco consultas y escríbanlas en cálculo o en algebra. [1]
[Escriban las sentencias en calculo y algebra. Si no lograron escribir alguna sentencia indiquen la razón y el punto de problema]

B) Seleccionen cinco consultas y escribanlas en calculo o en algebra

Consulta SQL

```
SELECT NAME
FROM world;
WHERE name = capital;
```

Algebra relacional

$\Pi_{name}(\sigma_{name=capital}(world))$

Consulta SQL

```
SELECT yr, subject, winner
FROM nobel;
WHERE yr = 1950;
```

Algebra relacional

$\Pi_{yr, subject, winner}(\sigma_{yr=1950}(nobel))$

consulta sql

SELECT winner

FROM nobel

WHERE yr = 1962 AND subject = 'literature';

Algebra relacional

$\pi_{winner}(\sigma_{yr = 1962 \wedge subject = 'literature'}(nobel))$

consulta sql

SELECT name

FROM world

WHERE population >

(SELECT population FROM world WHERE name = 'Russia')

Algebra relacional

$\pi_{name}(\sigma_{world.population > R.population}(world \times \pi_{population}(\sigma_{name='Russia'}(world))))$

consula sql

SELECT *

FROM nobel

WHERE winner = 'EUGENE O'NEILL';

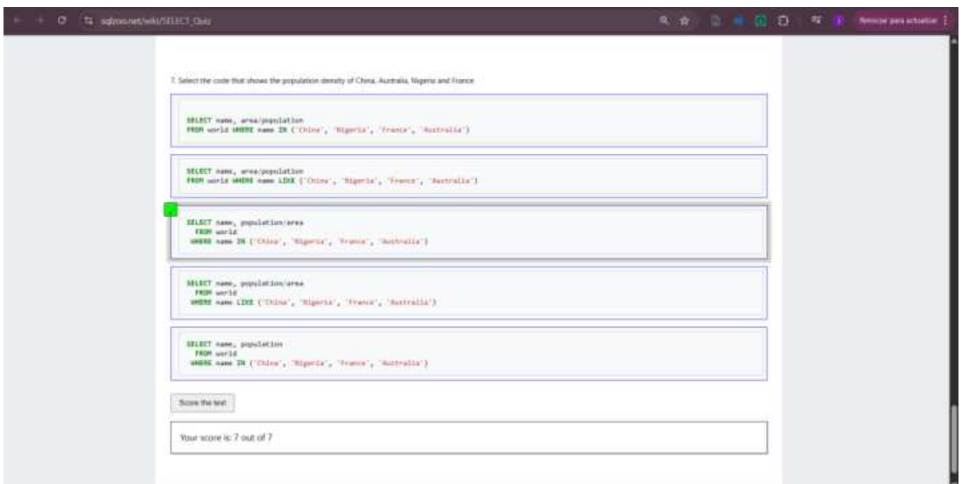
Algebra relacional

$\sigma_{winner = 'EUGENE\ O'NEILL'}(nobel)$

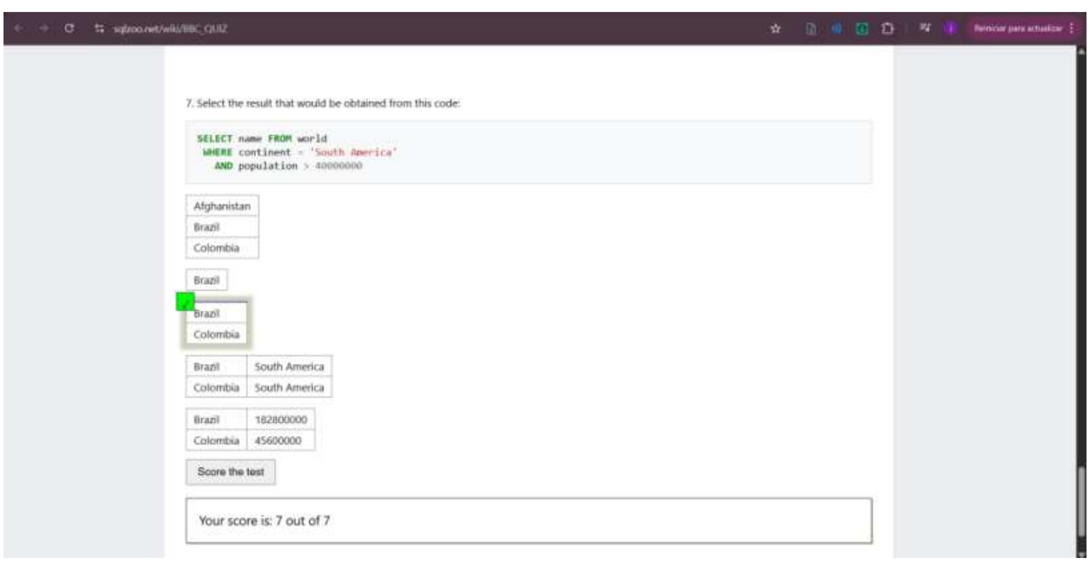
C. Presente los quices. Utilice el motor MySQL.

[Escriban el resultado obtenido de los quices]

SELECT Quiz



BBC QUIZ



Nobel Quiz

Nested SELECT Quiz

SUM and COUNT Quiz

sqlzoo.net/wiki/SUM_and_COUNT_Quiz

Africa	22550927
Asia-Pacific	28759578
Europe	23866987
North America	21660000

Table-C

Africa
Asia-Pacific
Europe
North America

Table-D

Americas	732240
Middle East	13403102
South America	17740392
South Asia	9437710

Table-E

Americas
Middle East
South America
South Asia

Score the test

Your score is: 8 out of 8

Usando SQLzoo.net [http://sqlzoo.net/]
[En auto01.doc]

A. Estudien la section `FUNCTIONS` de la referencia.

SQL Functions

Standard SQL functions and some common non-standard function. MS SQL, Oracle, MySQL and PostgreSQL provide support or have alternatives. Where possible alternatives are given.

ABS	CURRENT_TIMESTAMP	LENGTH	+ INTERVAL	SUBSTRING(ansi)
AVG	DATEPART	MAX	+ (string)	SUBSTRING
Strings	DAY	MIN	POSITION	SUM
CASE	DIV	MINUTE	QUARTER	TAN
CAST	EXTRACT	MOD	RANK	TO_CHAR(dates)
CEIL	FLOOR	MONTH	REPLACE	TRIM
COALESCE	HOUR	NULLIF	RIGHT	YEAR
CONCAT	IFNULL	NVL	ROUND	
COS	INSTR	PATINDEX	SECOND	
COUNT	LEFT	% MODULO	SIN	
CURRENT_DATE	LEN	+(dates)	SUBSTR	

```
booking(booking_id,booking_date,room_no,guest_id,occupants,room_type_requested,nights,arrival_time)
```

B. Proponga tres consultas que usen tres funciones diferentes.
Usen la tabla **booking** de Guest House. Utilice el motor My SQL.
[Escriban la consulta en lenguaje natural y la sentencia en SQL y ejecuten la sentencia SQL en sqlzoo. Si no lograron escribir alguna sentencia indiquen el punto de problema]

1.

SELECT COUNT(*) AS total_reservas

FROM booking;

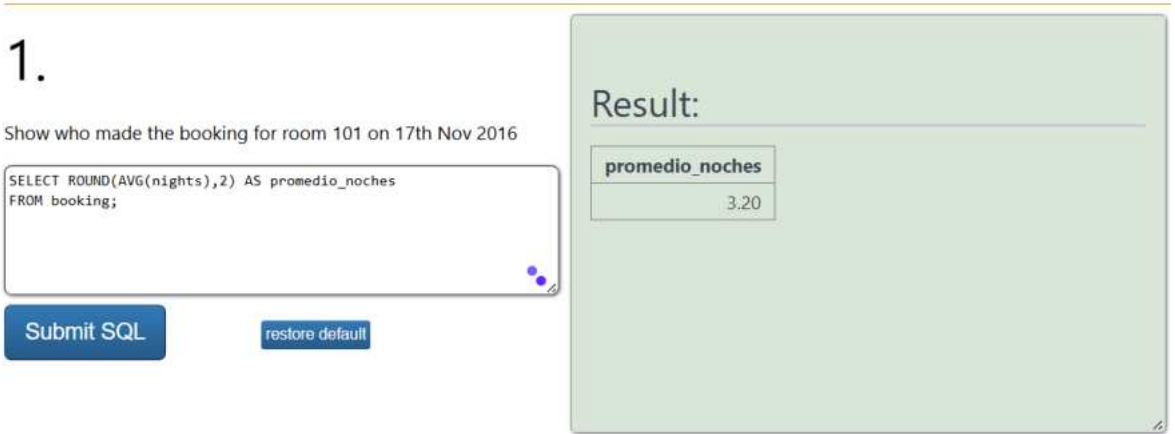
Lenguaje natural: “Contar cuántas reservas existen en total.”

2.

SELECT ROUND(AVG(nights),2) AS promedio_noches

FROM booking;

Lenguaje natural: “Calcular el promedio de noches reservadas.”



3.

SELECT *

FROM booking

WHERE YEAR(booking_date) = 2016;

Lenguaje natural: “Mostrar las reservas del año 2016.”

C. Propongan tres consultas una para cada esquema : 1) `GROUP BY ... HAVING …` **2)** `ORDER BY` 3) `DISTINCT`

Usen la tabla **booking** de Guest House. Utilice el motor My SQL.

[Escriban la consulta en lenguaje natural y la sentencia en SQL y ejecuten la sentencia SQL en sqlzoo. Si no lograron escribir alguna sentencia indiquen el punto de problema]

1.

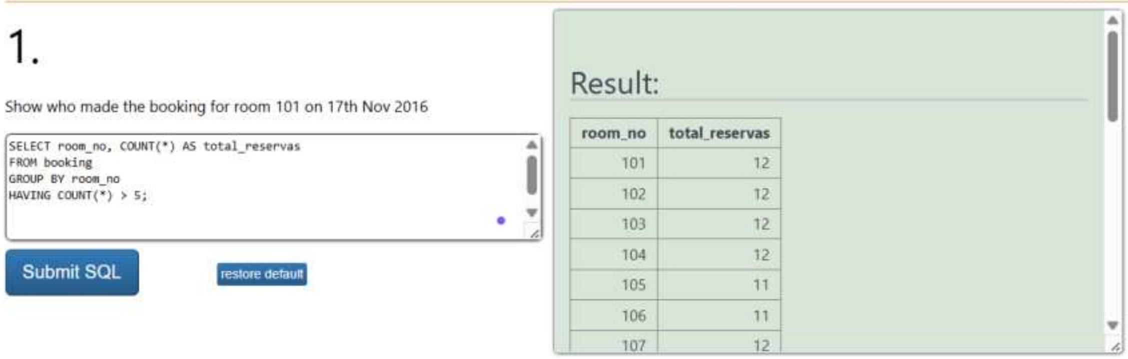
SELECT room_no, COUNT(*) AS total_reservas

FROM booking

GROUP BY room_no

HAVING COUNT(*) > 5;

Lenguaje natural: “Mostrar cuántas reservas tiene cada habitación, pero solo las habitaciones con más de 5 reservas.”



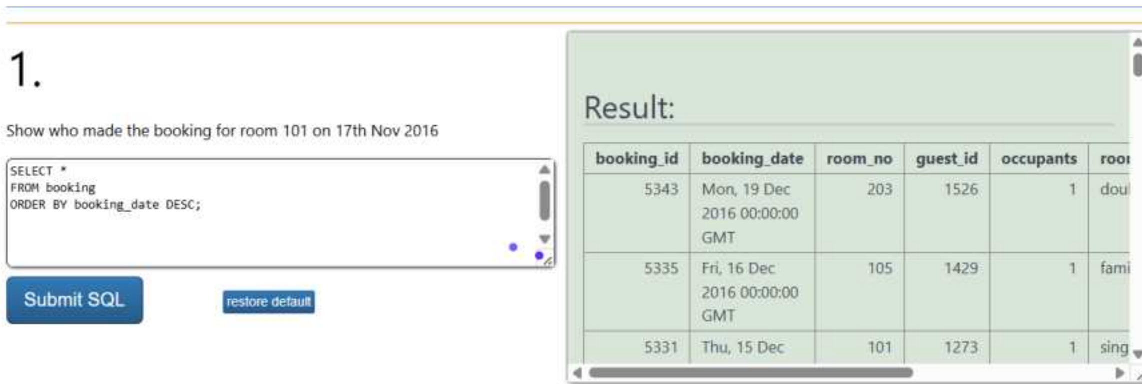
2.

SELECT *

FROM booking

ORDER BY booking_date DESC;

Lenguaje natural: “Mostrar las reservas ordenadas por fecha, de la más reciente a la más antigua.”



3.

SELECT DISTINCT room_type_requested

FROM booking;

Lenguaje natural: “Mostrar los tipos de habitación solicitados sin repetir.”